Marvyn Bailly

Curriculum Vitae

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RESEARCH INTERESTS

Mesh generation and grid adaptation, computational geometry, numerical optimization, numerical methods for partial differential equations, computational fluid dynamics, and scientific software that carries these methods from first principles to tools other researchers can use.

EDUCATION

University of Maryland, College Park: PhD, Applied Mathematics & Scientific Computation (GPA 3.89) *2023 - Present*

University of Washington: MS, Applied Mathematics, Numerical Methods (GPA 3.84) *2022 - 2023*

Oregon State University: BS, Mathematics, Applied & Computational (GPA 3.96) *2020 - 2022*

Graduate coursework: *Numerical Methods for Partial Differential Equations · Numerical Analysis · Numerical Optimization · Computational Fluid Dynamics · Scientific Computing · Parallel & High-Performance Computing · Natural Language Processing*

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Maryland *College Park, MD*

Computational Turbulence Lab, Dept. of Mechanical Engineering (Advisor: Dr. Johan Larsson) *2024 - Present*

- Research on grid generation and adaptation for wall-modeled turbulence simulation across Mach numbers, producing high-quality quad/hex grids for finite volume flow solvers.
- Designed mesh generation algorithms for numerical methods (FV/FD), from multi-block structured grids to metric-driven quad meshes, using computational geometry, discrete differential geometry, numerical optimization, and PDE-based methods to meet prescribed size, orientation, and smoothness.
- Developed these methods into documented, tested libraries in Julia, Python, and C++, with a GUI for non-experts, cutting mesh setup from days of manual work to hours of automatic computation.
- Ran the software on the Zaratan HPC cluster and presented the methods to multidisciplinary teams, translating simulation requirements into optimization objectives and constraints.

Dept. of Atmospheric & Oceanic Science (Advisor: Dr. Jacob Wenegrat) *2023 - 2024*

- Studied the effect of the diurnal cycle on internal ocean dynamics with a team of atmospheric and oceanic scientists.
- Extended a published ocean-convection study to a more realistic model built in Julia, running large eddy simulations on the Zaratan HPC cluster.
- Synthesized large simulation datasets into visualizations used to communicate results to the group.

PROFESSIONAL EXPERIENCE

Freelance Software Developer, Remote *Apr 2026 - Jul 2026*

Client: Neill Warrington (Postdoc, MIT) & Stephen Sorokanich (Quantum Fellow, Air Force Research Laboratory)

- Built Quantum Circuit Builder (JavaScript), a browser tool that turns superconducting circuit diagrams into Hamiltonian parameters for lattice field theory simulations, replacing a derivation previously done by hand.
- Delivered a drag-and-drop circuit editor, automatic extraction of the circuit Hamiltonian's parameters, and export in the format the clients' lattice simulation code consumes.
- Deployed it as a live web app for the clients to use directly in their research.

Fleet Attendant, Starship Technologies *Aug 2021 - Jun 2022*

Part-time, Corvallis, OR

- Troubleshot and resolved problems in the field across a fleet of autonomous sidewalk delivery robots.
- Worked hands-on with the robots, partner merchants, and customers to keep deliveries running.

TEACHING

Teaching Assistant, University of Maryland & University of Washington *2022 - Present*

- Led weekly discussion sections and problem-solving sessions, and gave feedback on student work, in graduate courses in data science and undergraduate courses in probability and calculus.

LEADERSHIP & SERVICE

Summer Camp Director, Girls Talk Math, University of Maryland *2023 - 2025*

- Co-directed the camp: ran program operations, hired undergraduate leaders, and coordinated industry outreach.

Graduate Student Committee, University of Maryland *2025 - 2026*

- Co-organized and led the weekly Joint AMSC, Math & Stat Student Seminar (JAMS).

AWARDS & HONORS

- Finalist, UMD Agentic AI Challenge (top 5 of 67 teams) *2026*

SELECTED PROJECTS

- **MetricRSP** (Python) — automatic quad-mesh generator for finite volume methods, where cell quality is critical: a target metric field in, a high-quality mesh out, via Delaunay background triangulation, cross-field design, an L-BFGS/Newton fold-free parameterization, and integer-grid quantization.
- **GridGeneration.jl** (Julia) — semi-automatic multi-block grid generator using algebraic and PDE-driven methods; ships generated API documentation and worked airfoil examples.
- **Smart Shelf Auditor** (Python) — team-built AI agent that audits retail shelves with object detection and a vision-language model; finalist, UMD Agentic AI Challenge (Apr 2026).
- **Mechanistic Interpretability of Grammatical Structures** (Python, PyTorch) — path patching across GPT-2 small's 12 layers and 144 attention heads to locate a name-resolution circuit.

TECHNICAL SKILLS

- **Programming:** Julia, Python, JavaScript, C++, MATLAB, Java.
- **Numerics & CFD:** finite volume/element methods, large eddy simulation, numerical optimization, SU2, Gmsh.
- **Tools:** Linux, Git, Make, SLURM, MPI, OpenMP, NumPy/SciPy, PyTorch, LaTeX.